$$x(t) = \sum_{n=-\infty}^{\infty} a_n e^{j\frac{2\pi nt}{T}} \quad \text{---} \quad (1)$$

$$x\left(t - \frac{T}{2}\right) = -x(t)$$

Put $t = t - \frac{T}{2}$ in (1)

$$\Rightarrow x\left(t - \frac{T}{2}\right) = \sum_{-\infty}^{\infty} a_n e^{j\frac{2\pi n}{T}\left(t - \frac{T}{2}\right)} = -x(t)$$

$$\Rightarrow \sum_{-\infty}^{\infty} a_n (-1)^n e^{j\frac{2\pi nt}{T}} = -\sum_{-\infty}^{\infty} a_n e^{j\frac{2\pi nt}{T}}$$

Comparing :

$$\Rightarrow a_n (-1)^n = -a_n$$

$$\Rightarrow a_n \left((-1)^n + 1\right) = 0$$

If n is even

$\Rightarrow a_n = 0$

$\Rightarrow a_{2m} = 0 \quad (\because n = 2m)$

If n is odd

$\Rightarrow a_n \cdot 0 = 0$

$\Rightarrow a_n$ can be anything

for $n = 2m + 1$